# Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

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**Abstract**

More and more people turn to large language models for mental health support. The models respond fluently, but ask the same question twice and you may get two different answers. In therapy, where trust depends on predictable guidance, that variation changes the treatment a patient receives.

This paper measures response consistency in the rescripting phase of Imagery Rehearsal Therapy, where a model must choose a therapeutic strategy and carry it out. Twelve models, spanning the EU-sovereign Mistral family, two proprietary ceilings (GPT-5.4, Claude Sonnet 4.6), and open-weight comparators, are evaluated across 7,200 trials at five temperatures. Every trial sees a byte-identical frozen dialogue history, so any variation comes from the model alone. A three-level framework compares strategy selections (Jaccard similarity), response texts (BERTScore), and whether the model does what it declared (LLM-judge alignment, $\kappa = 0.777$ against human annotation).

The main findings are: (1) Consistency is a model property, not a patient property. Median decision consistency spans 0.69 to 1.00 across models but barely moves across patient profiles. (2) The instability sits in strategy selection, not execution. Temperature changes which strategies a model picks, not how well it carries them out. (3) Greedy decoding is not optimal for all models. Ten of twelve reach their best mean consistency at a low non-zero temperature. (4) BERTScore detects when the tone changes but not when the strategy does, so no single metric shows the full pattern.

After the main collection, Mistral Medium 3.5 (the deployment's successor) and Command A (a dense control) were added, and GPT-5.4 was re-collected natively after its flat profile exposed a gateway silently dropping temperature. Medium 3.5 tops the stability ranking, every substantive finding survives, and the framework transfers to other protocols.

## 1 Introduction

Recurring nightmares are a common clinical problem with an effective treatment. Nightmare disorder affects roughly 4% of adults (Sandman et al., 2013; Li et al., 2010), and Imagery Rehearsal Therapy (IRT) is its best-evidenced protocol: the patient rescripts the nightmare under therapist guidance (Krakow et al., 2001; Morgenthaler et al., 2018). A persistent shortage of trained therapists limits access (Stade et al., 2024), making scalable, AI-assisted delivery a practical goal. Our group demonstrated the efficacy of a web-based IRT self-help intervention in a randomised controlled trial (Voss et al., 2026), and its successor, reDreamAI, is a consumer-facing IRT chatbot that guides users through the full protocol. It is not a research prototype but a real deployment target where reliability is non-negotiable, and it is the deployment context for this study.

The deployment context creates a specific problem. A general-purpose assistant has no therapeutic obligation, and response variability is acceptable or even desirable. A protocol-driven agent must make consistent strategic decisions. The rescripting step requires an active choice of therapeutic strategy, and an agent that selects different strategies on repeated runs of the same case is clinically unreliable, regardless of how capable any single response may be. Unlike rule-based chatbots, whose behaviour can be reviewed once before deployment, an LLM agent generates a new response on every run. Each run is a new draw from the output distribution, so stability must be evaluated repeatedly rather than verified once. The regulatory setting sharpens the requirement: the EU AI Act classifies autonomous therapeutic AI as high-risk and demands auditability (van Kolfschooten and van Oirschot, 2024; Aboy et al., 2024), and data sovereignty requirements push deployments toward self-hostable, EU-resident models (Dale, 2025). This constraint shapes the model selection in Section 4.1.

Existing evaluations do not measure this property. Standard benchmarks measure what a model *can* do, not whether it does the same thing reliably: a model can score well on a single evaluation while producing different therapeutic strategies on repeated runs of the same input. Capability and stability are distinct properties, but current evaluations conflate them. Clinical-LLM frameworks assess quality, safety, or efficacy of single outputs (Zhang et al., 2024; Chiu et al., 2024), so they cannot detect cross-run inconsistency. The consistency literature, meanwhile, studies repeated runs in other settings, such as factual answers (Novikova et al., 2025), creative diversity (Xu et al., 2025), numerical nondeterminism of inference (Yuan et al., 2025), and temperature effects on problem solving (Renze, 2024), but not the setting where inconsistency is clinically consequential: repeated *therapeutic decisions* within a structured protocol, where the decision space is a small set of clinically grounded strategies and a switch between them changes the treatment a patient receives.

This paper addresses the following research question:

> How consistent are LLM therapeutic responses across independent runs when given the same patient context within a structured IRT protocol?

Models declare their therapeutic strategy in a structured `<plan>` block before responding, which makes the clinical decision itself observable. Three contributions follow.

**Contribution 1.** A three-level evaluation framework that measures stability at each layer of the generation process: *plan consistency* (do strategy selections repeat?), *response similarity* (do the generated texts convey the same meaning?), and *plan–response alignment* (does the model do what it declared?). Together, the layers reveal divergence patterns that no single metric can detect.

**Contribution 2.** A cross-model comparison applying this framework to 12 models across 6 patient vignettes, 5 temperatures, and 20 trials per condition (7,200 trials), spanning EU-sovereign, proprietary, and open-weight model classes. Decision stability separates models sharply (median Jaccard 0.69–1.00) and is a model property rather than a patient-difficulty property. Response wording drifts with temperature faster than the underlying strategy choice.

**Contribution 3.** An extensibility demonstration that doubles as a robustness check: Mistral Medium 3.5 was added four months after the main runs, on identical frozen histories, with a single configuration entry. All substantive findings survive the model-set enlargement.

The study isolates the IRT rescripting phase because it is the stage that requires active strategy decisions. Isolating a single stage eliminates stage-routing confounds and allows any measured instability to be attributed to the therapist model rather than to upstream variation in the conversation.

## 2 Related Work

### 2.1 IRT and Digital Delivery

Imagery Rehearsal Therapy is the first-line psychological treatment for nightmare disorder (Krakow et al., 1995, 2001; Morgenthaler et al., 2018), with rescripting as its cognitive core (Albanese et al., 2022). Conversational agents in psychotherapy date back to ELIZA's Rogerian script (Weizenbaum, 1966). Modern digital and conversational delivery of structured protocols has shown feasibility and efficacy (Fitzpatrick et al., 2017; Heinz et al., 2025; Wu et al., 2026), including the randomised controlled trial of web-based IRT that precedes this work (Voss et al., 2026). Llama 3.3 70B, the model of that prior study, is included in the comparison for continuity. The present work asks the evaluation question that deployment raises: whether the LLM component of such a system makes stable clinical decisions.

### 2.2 Evaluating Clinical LLMs

Existing frameworks target competence and safety of single responses (Zhang et al., 2024; Iftikhar et al., 2025; Stade et al., 2024). The closest existing work, BOLT (Chiu et al., 2024), measures whether a model applies appropriate therapeutic techniques, but like its peers it evaluates single responses rather than the distribution across repeated runs. Protocol fidelity instruments from clinical psychology (Mowbray et al., 2003; Bellg et al., 2004; Kohrt et al., 2015) inspire the ternary alignment rubric used here. The focus of this study differs from all of these: not whether one response is good, but whether repeated responses agree.

### 2.3 LLM Output Consistency

Sampling temperature and stochastic decoding measurably change task behaviour (Renze, 2024), and even nominally deterministic inference carries numerical nondeterminism (Yuan et al., 2025). Novikova et al. (2025) survey the consistency landscape, largely over factual and task-answer settings, and Xu et al. (2025) quantify diversity collapse in creative generation. The present setting differs on both axes. The unit of consistency is a clinical decision drawn from a fixed therapeutic taxonomy, not a fact or a plot, and the context is a mid-protocol therapy state where run-to-run disagreement directly changes treatment. To our knowledge, decision-level stability within a structured clinical protocol has not been measured.

### 2.4 Chain-of-Thought and Declared Intent

Chain-of-thought prompting (Wei et al., 2022) structures generation, but CoT explanations are often unfaithful to internal computation (Turpin et al., 2023; Lanham et al., 2023). The plan block is therefore framed as declared intent rather than as a reasoning trace (Section 3.5). LLM-as-judge protocols (Zheng et al., 2023) underpin Method 3, with cross-family judging to mitigate self-preference bias.

## 3 Method

### 3.1 Two-Stack Architecture

The evaluation system separates dialogue creation from stability measurement through two decoupled stacks (Figure 1). The *generation stack* produces synthetic therapy sessions by running a multi-agent

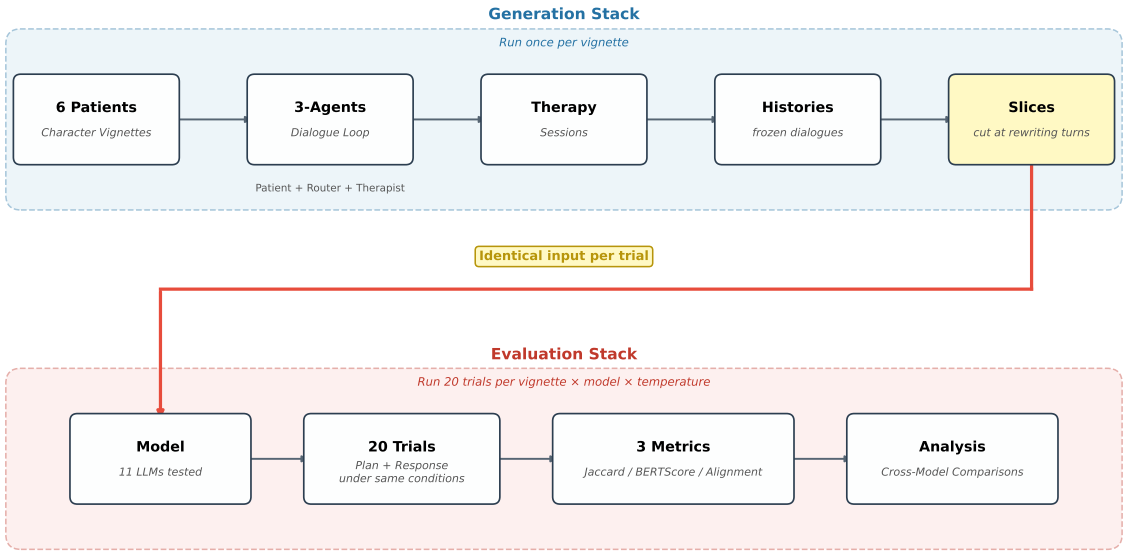


Figure 1: Two-stack pipeline. The generation stack produces frozen conversation histories (run once). The evaluation stack presents each history to the model under test across twenty independent trials per condition. The stacks are decoupled so that evaluation-side variance cannot be contaminated by generation-side randomness.

dialogue loop. The *evaluation stack* measures the stability of therapeutic responses by presenting frozen conversation histories to the model under test and collecting multiple independent outputs.

This separation is a deliberate design choice. If the evaluation stack generated its own dialogue context, any measured instability could reflect upstream randomness in the patient or routing agents rather than genuine variation in the therapist model. By freezing the conversation history before evaluation begins, every trial receives identical input, and any observed inconsistency is attributable solely to the model under test.

### 3.2 Dialogue Generation

The generation stack orchestrates a three-agent loop that produces naturalistic IRT sessions. Each turn passes through a *patient* agent that responds based on a behavioural profile (nightmare content, emotional style, engagement pattern), a *router* agent that classifies the current protocol stage, and a *therapist* agent that generates a stage-appropriate response. The loop traverses the full IRT protocol (Figure 2): the dialogue cycles through recording and rescripting until each stage is complete, then proceeds to rehearsal, and the complete session is exported. Frozen history slices for the experiment are cut from the rescripting stage.

Six patient vignettes cover the range of clinical presentation difficulty expected in IRT deployment: *anxious*, *avoidant*, *cooperative*, *resistant*, *skeptic*, and *trauma*. They range from a cooperative patient who engages readily with rescripting instructions to a trauma patient whose nightmare content triggers intense emotional responses mid-session. The vignette design is informed by anonymised

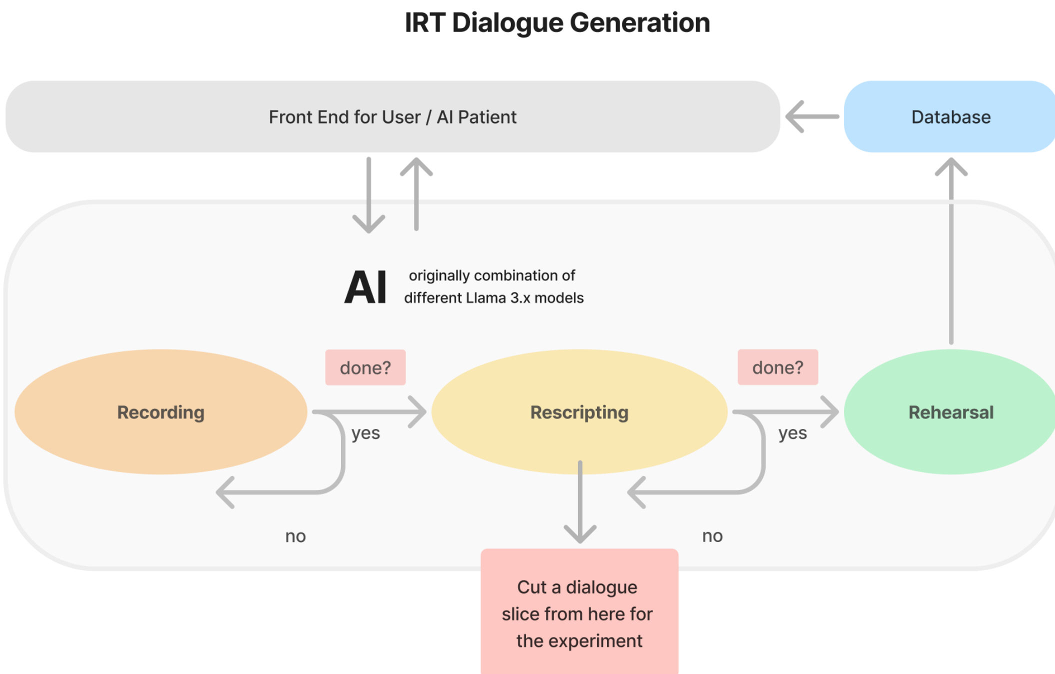


Figure 2: Dialogue generation loop. The three-agent system traverses recording, rescripting, and rehearsal. Frozen history slices for the experiment are cut at rescripting-turn boundaries.

session data from the prior reDreamAI efficacy study; data protection requirements prevent using real transcripts directly, so synthetic profiles were created and validated against published work on simulated patients. Wang et al. (2024b) demonstrate that expert-defined behavioural principles produce clinically plausible simulated patients, and Wang et al. (2024a) provide conversational styles (direct, resistant, expressive, minimal, deviating, agreeable) that parallel the six profiles used here. Six profiles sample the difficulty range but do not claim exhaustive coverage of patient variability.

### 3.3 Frozen Histories and Conversation Depth

Frozen conversation histories serve as deterministic evaluation entry points. Each vignette's session is sliced at rescripting-turn boundaries, where each slice ends after a complete therapist–patient exchange. Every trial, across all models and temperatures, receives the identical frozen history as input.

The evaluation depth was chosen empirically. A preliminary analysis tested Mistral Large 3 at five slice depths across all six vignettes: consistency was lowest at the second slice (mean Jaccard 0.77 at $T$=0.075), where the strategy is not yet anchored by prior exchanges, and rose to near-ceiling values from the third slice onward ($J > 0.93$). The main experiment therefore evaluates at the second rescripting boundary (slice 2), making all reported stability scores a conservative lower bound: a model that is consistent here would likely be at least as consistent later in the conversation.

### 3.4 Evaluation Stack and Fused Generation

The evaluation stack bypasses the router and injects the rescripting prompt directly. Because the protocol stage is known a priori (all slices lie within the rescripting phase), routing is unnecessary and

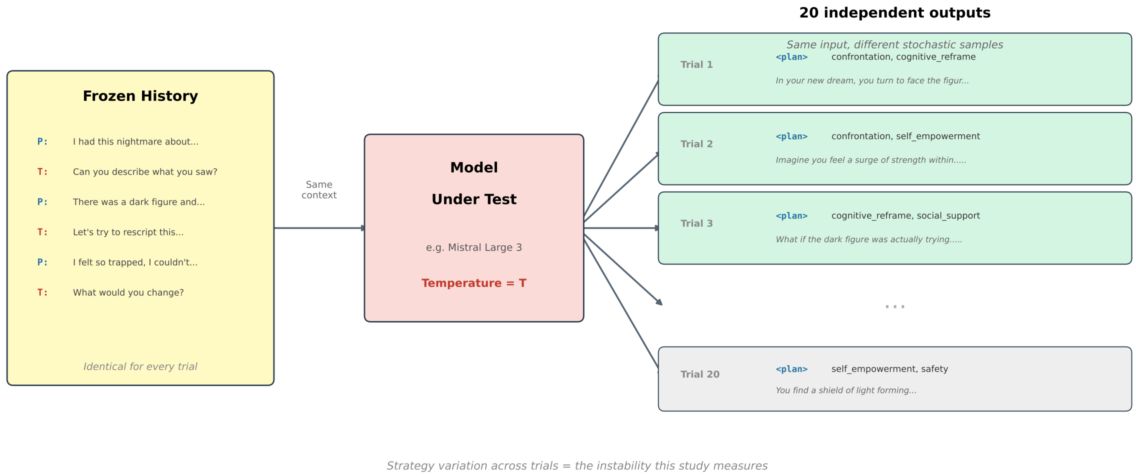


Figure 3: Measurement design. The same frozen history is presented to the model under test twenty times. Each trial independently produces a plan and a therapeutic response. Variation across trials is the instability this study measures.

would only add a source of variation. Bypassing it isolates the therapist model from stage-classification error.

Each trial uses *fused* generation: the model produces a plan block, <plan>strategy_1 / strategy_2</plan>, declaring one or two therapeutic strategies from a fixed taxonomy, followed by its therapeutic response, in a single call. The plan tokens condition the response tokens in a chain-of-thought style generation. This fusion is what makes the alignment analysis of Method 3 meaningful: because plan and response come from a single pass, any mismatch between declared intent and executed strategy reflects the model's own inconsistency rather than a decoupling artefact. Figure 3 illustrates the measurement design.

Twenty independent trials are collected per condition, yielding $\binom{20}{2} = 190$ pairwise comparisons. This trial count balances statistical coverage of the pairwise similarity distribution against compute cost.

## 3.5 The Plan Mechanism

The plan block raises a methodological question. The faithfulness literature shows that chain-of-thought explanations do not always reflect a model's actual computation (Turpin et al., 2023; Lanham et al., 2023), so a declared strategy may be a post-hoc rationalisation rather than a window into decision-making. This study sidesteps the problem by framing the plan as *declared intent*: the plan captures what the model commits to, not what it internally computes. Whether or not the declaration reflects genuine reasoning, a practical consequence holds: if strategy declarations vary across runs, clinical responses will also vary.

The framing preserves three uses. The declared strategies enable consistency scoring at the decision level (Method 1). Comparing declaration against response enables alignment verification (Method 3). And the plan provides a human-readable summary of the model's intended approach, which is what a clinical overseer would review in deployment.

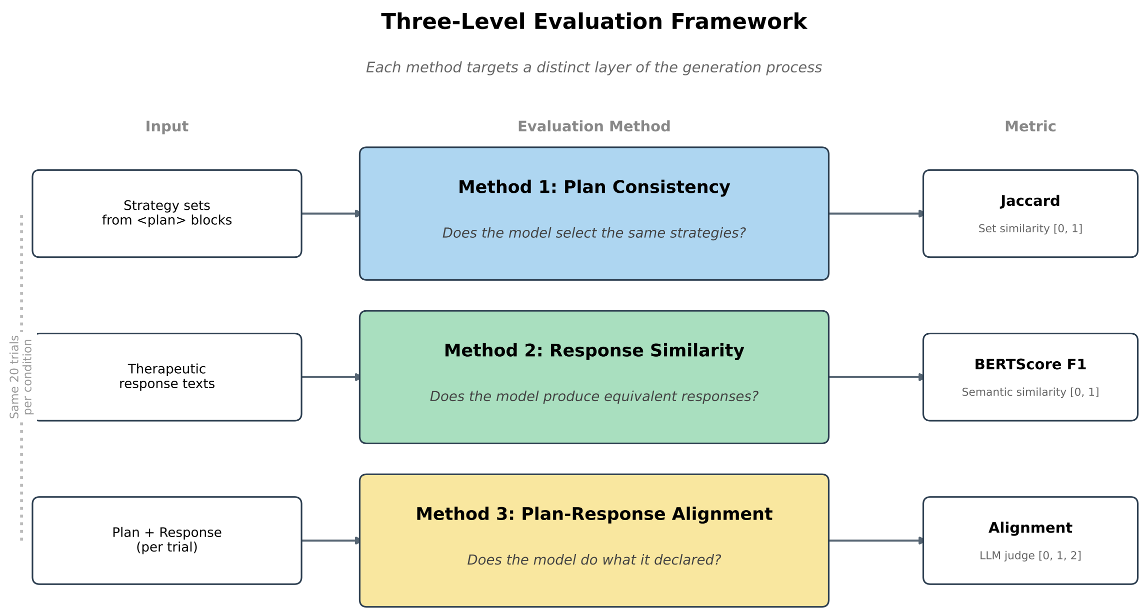


Figure 4: Three-level evaluation framework. Each method targets a distinct layer of the generation process: strategy selection (Jaccard), response wording (BERTScore), and plan–response alignment (LLM judge).

## 3.6 Strategy Taxonomy

A fixed taxonomy constrains the plan block to a predefined strategy set. The constraint is necessary for quantitative evaluation: free-text plans cannot be aggregated into set-similarity scores. Declarations are limited to six categories describing concrete rescripting *mechanisms*, not therapeutic *goals*: `confrontation` (external action toward the threat), `self_empowerment` (internal transformation), `safety` (environment modification), `cognitive_reframe` (meaning change), `social_support` (adding allies), and `sensory_modulation` (sensory and calming details).

The taxonomy went through four iterations because early goal-level labels produced extreme skew: a single broad "empowerment" category absorbed 87.7% of all picks, since every rescripting intervention serves the goal of increased mastery. Splitting along the mechanism dimension resolved the skew (Appendix A). Avoidance and distancing are deliberately excluded: IRT is understood to work by increasing mastery over the nightmare rather than by avoiding it (Albanese et al., 2022), and rescripted-dream content relates to treatment outcome (Harb et al., 2012). Taxonomy design directly shapes measured consistency, a dependence revisited in Section 6.5.

## 3.7 Three Evaluation Metrics

Three metrics target distinct layers of the generation process (Figure 4). Method 1 asks whether the model makes consistent therapeutic decisions. Method 2 asks whether it produces consistent therapeutic responses. Method 3 asks whether it does what it says it will do.

### 3.7.1 Method 1: Plan Consistency

Strategy sets are extracted from the plan block of each trial and compared with pairwise Jaccard similarity, $J(A, B) = |A \cap B|/|A \cup B|$, averaged over all 190 trial pairs per condition. Jaccard is appropriate because strategy combinations are unordered sets: a model that declares *confrontation / cognitive reframe* has made the same clinical decision as one that declares *cognitive reframe / confrontation*. Because the sets contain one or two items drawn from six categories, Jaccard takes only a small number of distinct values (0, 0.33, 0.5, 0.67, 1.0), which motivates the rank-based statistics in Section 4.3. A plan *validity rate* (fraction of trials whose plan block parses to taxonomy categories) serves as an upstream quality gate.

### 3.7.2 Method 2: Response Similarity

Pairwise BERTScore F1 (Zhang et al., 2020) is computed over the response texts (plan blocks excluded) across the same 190 pairs, with DeBERTa-XLarge-MNLI (He et al., 2021) as the embedding model. It ranks first of 130+ models on WMT16 human correlation ($r = 0.778$), and its NLI fine-tuning matches the semantic-equivalence judgement the metric requires: surface metrics such as BLEU or ROUGE would penalise valid therapeutic paraphrasing.

### 3.7.3 Method 3: Plan–Response Alignment

An LLM judge scores each declared strategy on a ternary scale: 0 (absent), 1 (partial), 2 (implemented), normalised to $[0, 1]$ per trial. The ternary scale follows clinical fidelity frameworks (ENACT, NIH Behavior Change Consortium), where partial implementation is a clinically meaningful state that binary scoring would lose. The judge is Gemini 3 Flash at $T$=1.0 (the Gemini 3 default, which Google advises against lowering), with suspect trials re-judged by Gemini 3.1 Pro. Bias mitigations follow established recommendations (Zheng et al., 2023): cross-family judging (no Gemini model appears among the evaluation targets), chain-of-thought justification, and a transparent rubric (Appendix C). NLI cross-encoders were considered and rejected after preliminary testing showed an F1 ceiling near 0.55 on this task.

Judge reliability was validated by human annotation of a stratified random subset of 60 trials scored on the same rubric. Linear-weighted Cohen's $\kappa$ between human and judge is 0.777 (substantial agreement), with 89.2% exact agreement. All 13 disagreements go in one direction, the human scoring higher than the judge, indicating a conservative rather than lenient judge. Judged trials that fail (timeout or API error) or are unjudgeable (invalid plan) are treated as missing data and excluded from the alignment mean rather than scored zero; invalid plans are already penalised by the validity rate.

Alignment primarily serves as a validation check: scores near the ceiling confirm that models implement what they declare, validating Jaccard as a measure of genuine decision stability rather than label noise.

# 4 Experimental Setup

## 4.1 Models

EU data sovereignty is the primary selection criterion for the study's main subject: the reDreamAI deployment context requires a model that can be self-hosted within EU data residency boundaries (Dale, 2025). At design time that pointed to Mistral Large 3, the EU-sovereign Apache-2.0 flagship. Around it, the comparison set (Table 1) spans a size-class ladder from 24–32B dense models through

| Model | Size | Class | License |
|---|---|---|---|
| Mistral Small 3.2 | 24B dense | small | Apache 2.0 |
| Mistral Small 4 | 119B MoE (6.5B act.) | large MoE | Apache 2.0 |
| Mistral Medium 3.5 | 128B dense | large dense | modified MIT |
| Mistral Large 3 | 675B MoE (41B act.) | large MoE | Apache 2.0 |
| Qwen 3.5 27B | 27B dense | small | Apache 2.0 |
| Qwen 3.5 122B | 122B MoE (10B act.) | large MoE | Apache 2.0 |
| Qwen 3.5 397B | 397B MoE (17B act.) | large MoE | Apache 2.0 |
| OLMo 3.1 32B | 32B dense | small | Apache 2.0 (full provenance) |
| Llama 3.3 70B | 70B dense | mid | Llama license |
| Command A | 111B dense | large dense | CC-BY-NC |
| GPT-5.4 | undisclosed | proprietary | — |
| Claude Sonnet 4.6 | undisclosed | proprietary | — |

Table 1: Evaluation targets (therapist role). All models run in non-thinking mode and are accessed via OpenRouter, except GPT-5.4 (native OpenAI API, Section 4.1). Supporting roles are fixed across all runs: patient = Dolphin Mistral Venice 24B, router = Llama 3.3 70B, judge = Gemini 3 Flash with Gemini 3.1 Pro fallback (no family overlap with any evaluation target).

70B dense and 119–675B MoE up to two proprietary ceilings. Three scales of Qwen 3.5 probe within-family scaling, OLMo 3.1 provides full training-data provenance, and Llama 3.3 70B maintains continuity with the prior efficacy study. All models run in non-thinking mode; where a model exposes a configurable reasoning mode, it is explicitly disabled. Mistral Medium 3.5, added after the main collection (below), ultimately overtakes Large 3 as the recommended EU-sovereign candidate (Section 6.2).

Mistral Medium 3.5 (128B dense, open weights under a modified MIT license, released 2026-04-30) was added after the main data collection, using the identical frozen histories, prompts, and analysis pipeline. The addition is deployment-driven, not incidental: Medium 3.5 has since been selected as the successor model for the reDreamAI chat deployment, and its dense architecture raised the expectation of higher run-to-run stability, since a dense forward pass has no expert routing to vary between runs. The addition also serves two methodological purposes: it tests whether the framework extends to a new model with a single configuration entry, and it tests whether the main findings are robust to enlarging the model set (Section 6.4).

Command A (Cohere, 111B dense, no reasoning variant) was later added by the same single-entry route as a control for the dense-architecture expectation: a second model in the 111–128B dense weight class, from a different vendor, shows whether stability tracks the architecture class or the specific model. All models are accessed through the OpenRouter gateway except GPT-5.4. End-to-end tests of parameter forwarding caught OpenRouter accepting but silently dropping `temperature` for GPT-5.1 and later models, so GPT-5.4 is accessed directly through the native OpenAI API, which validates sampling parameters rather than ignoring them, and its earlier gateway runs are excluded from all analyses (Section 5.7).

### 4.2 Conditions

The full design crosses 12 models × 6 vignettes × 5 temperatures × 20 trials at slice 2, i.e. 600 trials per model and 7,200 trials in total.

The five-point temperature sweep, $T \in \{0.0, 0.075, 0.15, 0.3, 0.6\}$, is anchored in vendor documentation. $T$=0.0 (greedy decoding) serves as the deterministic reference, although GPU parallelisation leaves residual floating-point nondeterminism even there (Yuan et al., 2025), which the twenty-trial design captures directly (Blackwell et al., 2024). $T$=0.15 is the Mistral vendor default. $T$=0.6 is the Qwen 3.5 and OLMo default, close to the standard Llama chat range and the proprietary defaults ($\approx 0.7$). $T$=0.3 fills the gap, and $T$=0.075 was added after an initial four-point run revealed that several models peak in consistency *between* $T$=0.0 and $T$=0.15, a pattern examined in Section 5.2.

No reproducibility seed was set, since not all vendors support one and even where available it is best effort at most. A supplementary experiment on Mistral Large 3 confirms this choice: fixing `seed=42` does not improve consistency over unseeded runs (Appendix F). Trial repetition captures the stochastic variance directly.

### 4.3 Statistical Analysis

The analysis is primarily descriptive, and all tests are exploratory rather than confirmatory: the study describes stability patterns, it does not test a preregistered hypothesis. Spearman rank correlations quantify per-model temperature sensitivity and cross-metric relations (rank-based, because Jaccard takes few discrete values and BERTScore distributions may be non-normal). Kruskal–Wallis $H$ assesses model and vignette differences, with pairwise Mann–Whitney $U$ under Bonferroni correction ($\binom{12}{2} = 66$ comparisons) and $\eta^2$ effect sizes alongside $p$-values.

No fixed threshold defines "stable enough"; clinical significance thresholds for LLM stability do not yet exist. Results are interpreted comparatively, model against model and across the temperature sweep, with the random Jaccard baseline (expected value for random 2-of-6 selections, $\approx 0.2$) as the lower reference point.

### 4.4 Reproducibility

Model endpoints are pinned by model IDs; exact IDs, prompts, and aggregated per-condition results are released with the code at `https://github.com/me-daniel/measure-ai-drift`. The main set was collected 2026-03-21 to 2026-03-24, Mistral Medium 3.5 on 2026-07-09, and Command A and the GPT-5.4 re-collection (same dated snapshot, `gpt-5.4-2026-03-05`) on 2026-07-28; a `run_date` column in the released data records the collection date of every run. The frozen histories and per-trial raw outputs are retained by the author and available on request; a public release is planned. The original collections used OpenRouter's default routing, under which the third-party serving stack (provider, quantisation) may differ between requests. The 2026-07-28 collections lock this down: each model is pinned to a single serving provider, requests are rejected rather than silently modified if an endpoint cannot honour a sampling parameter, and the serving provider is logged per trial. The judge endpoints are versioned previews. These caveats are discussed in Section 6.6.

## 5 Results

All results are reported at the second rescripting turn (slice 2), aggregated across six vignettes. Statistics are computed over 360 experiment runs (12 models $\times$ 6 vignettes $\times$ 5 temperatures, 20 trials each). Tests are exploratory. Kruskal–Wallis $H$ is reported with $\eta^2$ effect sizes, and pairwise contrasts use Mann–Whitney $U$ with Bonferroni correction over $\binom{12}{2} = 66$ comparisons.

### 5.1 Plan Validity and Strategy Distribution

Before analysing stability, the plan extraction step must be validated. The structured format turns out to be essentially free: overall validity is 99.8%, with every model at or above 98.3%

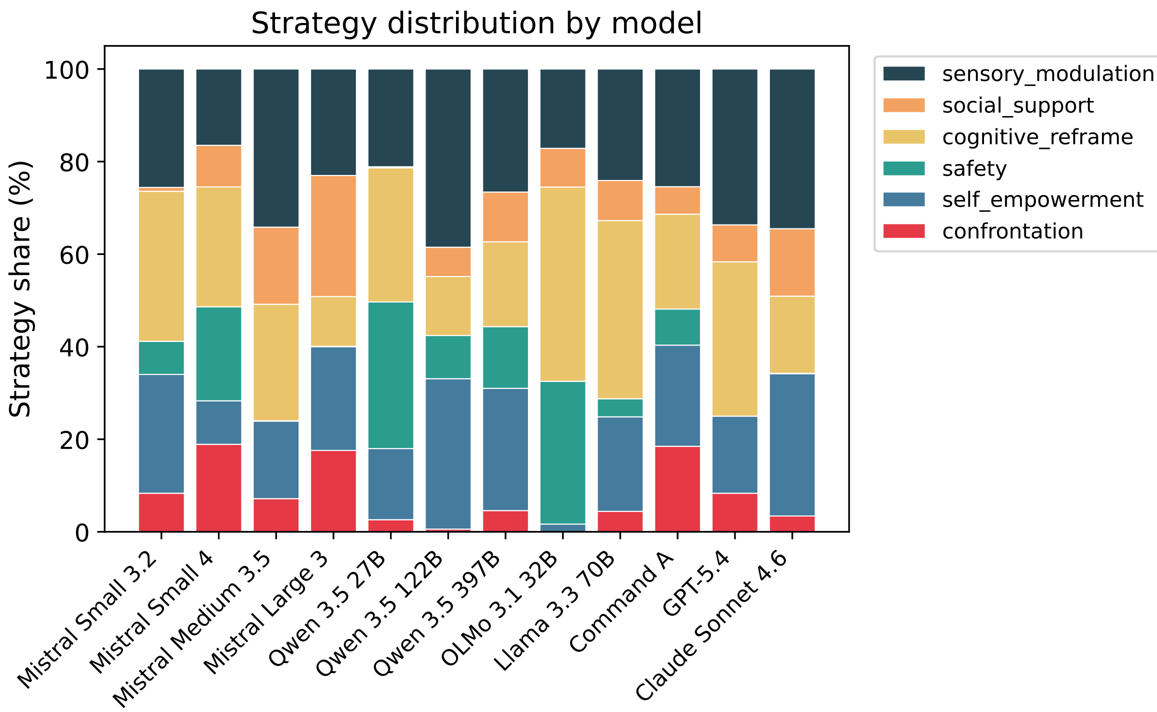


Figure 5: Strategy distribution by model, pooled across temperatures and vignettes. Each bar shows the share of trials selecting each taxonomy category. Models with narrower distributions have less room to vary.

(Command A lowest; Figure 11), and validity does not vary meaningfully with temperature. The fused prompt reliably elicits structured plan output, and the stability metrics below are computed over near-complete data. The seventeen malformed trials are treated as missing data (Section 3.7).

Across 14,361 declared strategy picks, all six taxonomy categories are used, though not uniformly: `sensory_modulation` (26.7%), `cognitive_reframe` (25.4%), and `self_empowerment` (20.0%) dominate, while `safety` (10.4%), `social_support` (9.6%), and `confrontation` (7.9%) form a long tail (Figure 5). No category exceeds a third of picks, in contrast to the 87.7% single-category skew of the initial taxonomy (Appendix A).

These distributional differences are a prerequisite for interpreting the consistency results that follow. A model that always selects the same two strategies will score high on Jaccard regardless of temperature, while a model with a wider repertoire has more room to vary. OLMo 3.1 32B and Qwen 3.5 122B concentrate most (73% and 71% of picks in their top two categories), the proprietary models follow (65–67%), and Command A, Mistral Large 3, and Mistral Small 4 spread widest (46–49%).

### 5.2 Plan Consistency (Method 1)

Figure 6 presents the headline result: decision-level stability is high but far from universal. Median pairwise Jaccard per run ranges from 1.000 (GPT-5.4, Mistral Medium 3.5, Mistral Small 3.2, OLMo 3.1 32B, Qwen 3.5 122B) down to 0.689–0.727 (Qwen 3.5 397B, Llama 3.3 70B, Qwen 3.5 27B). Even at fixed temperature and identical context, the least stable models change their declared therapeutic strategy in roughly one of three trial pairs.

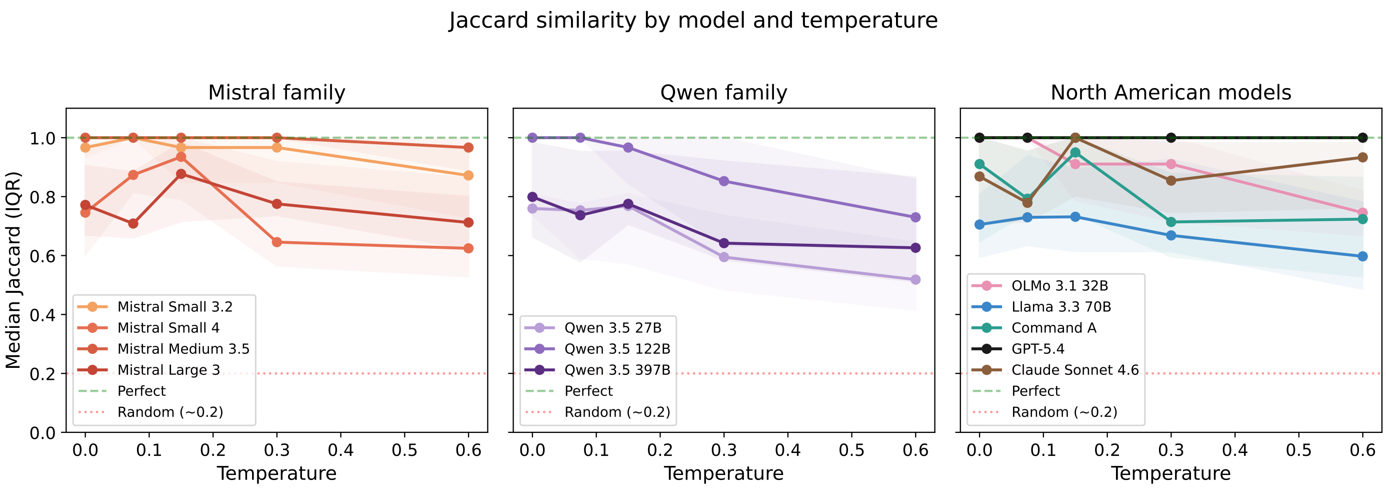


Figure 6: Median pairwise Jaccard by model and temperature, grouped by family. Shaded bands show IQR. The dashed line marks perfect consistency, the dotted line the random baseline ($J \approx 0.2$). Higher is more consistent.

Temperature degrades decision consistency overall (pooled Spearman $\rho = -0.215$, $p < .001$; Kruskal–Wallis $H(4) = 21.7$, $p < .001$, $\eta^2 = 0.060$): the grand median falls from 0.967 at $T$=0.0 to 0.747 at $T$=0.6. But the effect is far from uniform. Per-model correlations separate three groups: OLMo 3.1 32B ($\rho = -0.65$), Qwen 3.5 122B ($\rho = -0.56$), and Qwen 3.5 27B ($\rho = -0.49$) degrade significantly, the Mistral family and Command A show no significant trend, and the proprietary ceilings are flat ($\rho = +0.03$ for Sonnet 4.6, +0.09 for GPT-5.4). The most capable models hold their therapeutic decisions stable across the whole sampled range.

Greedy decoding, however, is not always optimal. Ten of the twelve models reach their best mean Jaccard at a non-zero temperature. The pattern is clearest in the Mistral family: Mistral Small 4 rises from 0.777 at $T$=0.0 to 0.881 at $T$=0.15, Mistral Large 3 from 0.796 to 0.848 at $T$=0.15, and Llama 3.3 70B from 0.719 to 0.775 at $T$=0.075. The post-hoc additions follow the same pattern: Mistral Medium 3.5 peaks at a perfect 1.000 mean at $T$=0.075, above its 0.979 at $T$=0.0, and Command A rises from 0.832 to 0.861 at $T$=0.15. Whether these differences reflect a genuine benefit of low sampling variance or sample variation ($n = 6$ vignettes per cell) cannot be determined from these results alone, but they are consistent with Mistral's vendor default of $T$=0.15.

Model differences are substantial (Kruskal–Wallis $H(11) = 104.5$, $p < .001$, $\eta^2 = 0.291$; 20 of 66 Bonferroni-corrected pairs significant). Modal-set agreement, the fraction of trials whose declared set equals the run's most common set, tells the same story (Figure 7), from 0.993 (GPT-5.4) and 0.972 (Mistral Medium 3.5) down to 0.670 (Qwen 3.5 27B).

### 5.3 Response Similarity (Method 2)

BERTScore F1 measures semantic similarity between the response texts, independent of the plan block. The pooled temperature effect is roughly twice as strong as for Jaccard ($\rho = -0.489$, $p < .001$; $H(4) = 93.6$, $p < .001$, $\eta^2 = 0.261$), with the median falling from 0.787 at $T$=0.0 to 0.705 at $T$=0.6 (Figure 8). Rising temperature changes response wording before it changes strategy selection: models say the same thing differently before they decide differently.

BERTScore is, however, insensitive to strategy changes. A model that switches from *confrontation* to *cognitive reframe* produces surface text similar enough that BERTScore barely registers the

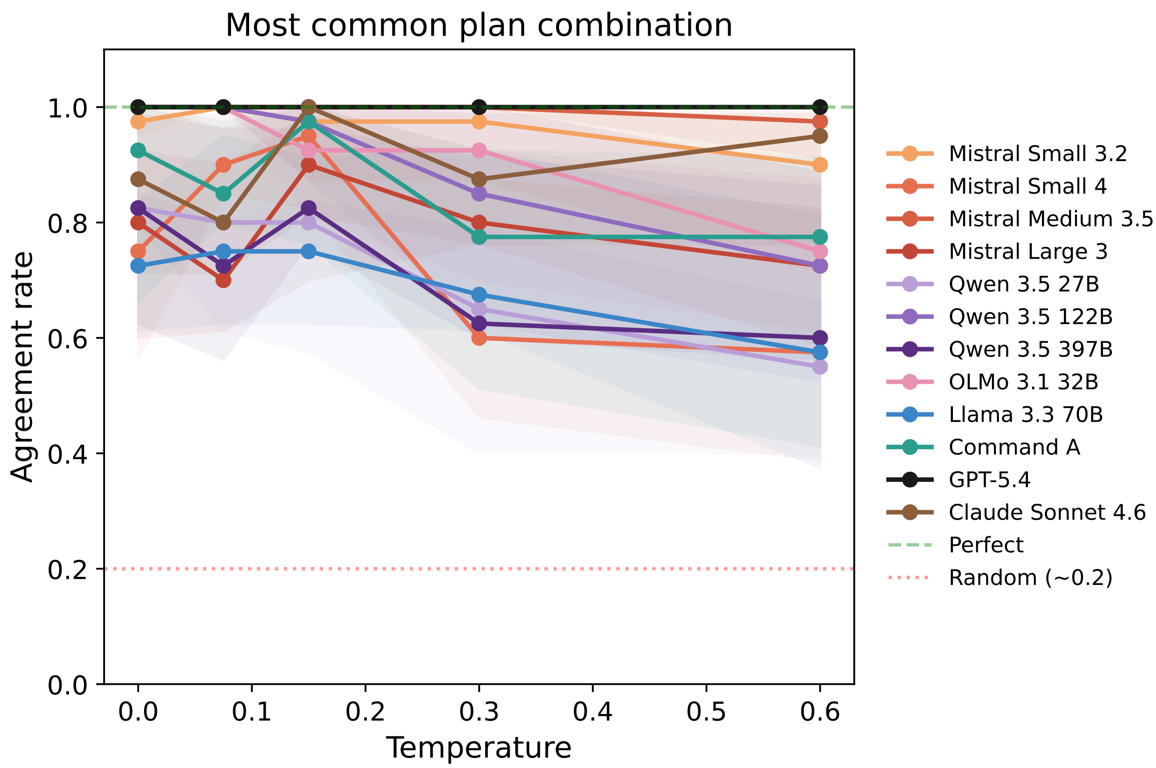


Figure 7: Modal-set agreement by model: fraction of trials whose declared strategy set equals the run's modal set.

difference, since different rescripting strategies share the same conversational framing, sentence length, and register. BERTScore therefore serves as a check on response style rather than as a measure of decision stability. Between-model differences are again large ($H(11) = 92.0$, $p < .001$, $\eta^2 = 0.256$; 19 of 66 pairs significant), with Mistral Medium 3.5 highest (mean F1 0.816) and Mistral Small 4 lowest (0.697).

### 5.4 Vignette Effects

Patient presentation matters little relative to model and temperature. Vignette differences are significant for neither decision consistency ($H(5) = 6.3$, $p = .280$, $\eta^2 = 0.018$) nor response similarity ($H(5) = 11.0$, $p = .052$, $\eta^2 = 0.031$), although the latter sits at the boundary: patient profile affects response wording more than strategy selection. Median Jaccard spans a narrow 0.818 (cooperative) to 1.000 (anxious, skeptic). Stability is predominantly a *model* property, not a *patient-difficulty* property, though individual model–vignette combinations can still deviate from the aggregate, which keeps per-vignette reporting useful for spotting where a specific model breaks down. The full per-vignette heatmaps appear in Appendix E.

### 5.5 Cross-Metric Relations

The three methods are related but not redundant. Decision- and response-level consistency correlate moderately (Spearman $\rho = 0.552$, $p < .001$, $n = 360$): models with more consistent strategy selection also tend to produce more similar text, but not strongly enough for one metric to substitute for the

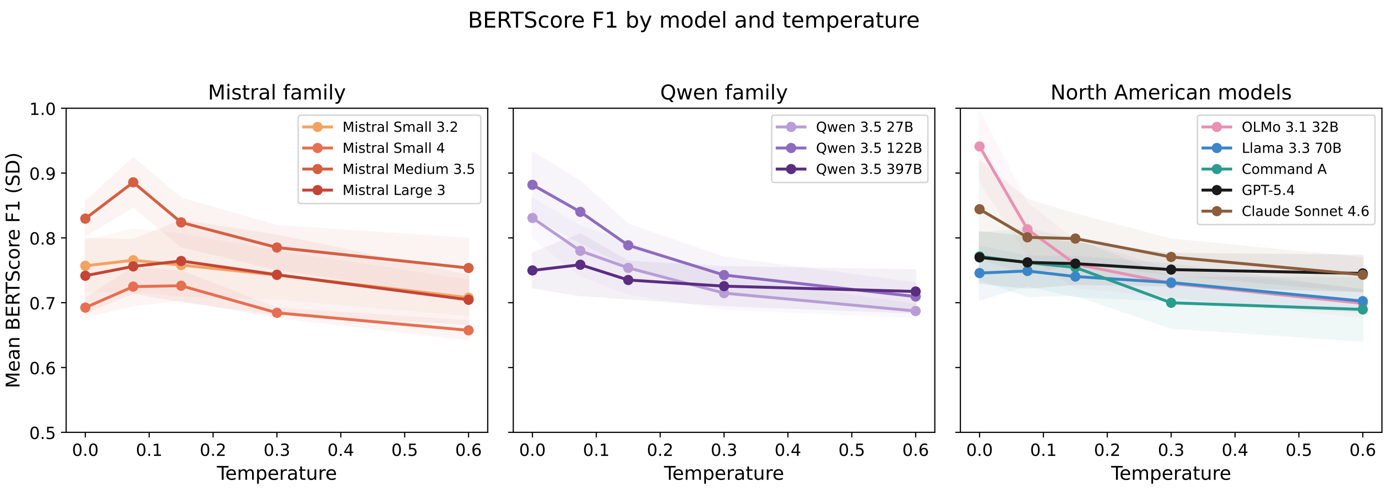


Figure 8: Pairwise BERTScore F1 by model and temperature, grouped by family.

other. Alignment correlates only weakly with Jaccard ($\rho = 0.293$) and moderately with BERTScore ($\rho = 0.403$), consistent with its role as a near-ceiling validation check rather than a third stability axis.

Figure 9 shows how the models arrange on the two stability axes. Mistral Medium 3.5 anchors the upper right, five models sit at perfect median decision consistency with clearly different response-level scores, and Qwen 3.5 27B occupies the lower left. Mistral Small 4 presents an interesting case: moderate decision consistency but the lowest response similarity in the set, so even when it selects the same strategies, its wording varies considerably.

## 5.6 Plan–Response Alignment (Method 3)

Models largely do what they declare. Mean alignment ranges from 1.000 (GPT-5.4 and Sonnet 4.6) down to 0.810 (Mistral Small 4), and it is temperature-independent ($\rho = -0.069$, $p = .19$). Models do not get worse at implementing their declared strategies as temperature rises: they select *different* strategies more often (lower Jaccard) but implement whichever strategy they select equally well (Figure 10).

Model differences on alignment are nonetheless the largest of the three metrics ($H(11) = 155.8$, $p < .001$, $\eta^2 = 0.434$; 34 of 66 pairs significant), driven by the gap between the proprietary ceilings (1.000) and the weaker open models. The near-ceiling scores validate Method 1: declared plans are implemented, so Jaccard measures genuine decision variability rather than label noise. Judge accounting: of 7,200 trials, none failed with judge errors, 17 were unjudgeable (invalid plans, 10 of them Command A's, excluded as missing data), and 9 trials (0.13%) were re-judged by the Gemini 3.1 Pro fallback.

## 5.7 Robustness to Model-Set Changes

The first post-hoc addition lands at the top of the stability ranking. Mistral Medium 3.5 attains a perfect median Jaccard of 1.000, the highest mean BERTScore F1 in the set (0.816), the second-highest modal-set agreement (0.972), perfect plan validity, and mean alignment of 0.961 across its 30 conditions. Its Jaccard profile stays at or near ceiling across all five temperatures. This ceiling is not the sampling artefact diagnosed for GPT-5.4 below: Medium 3.5's BERTScore falls by 0.13

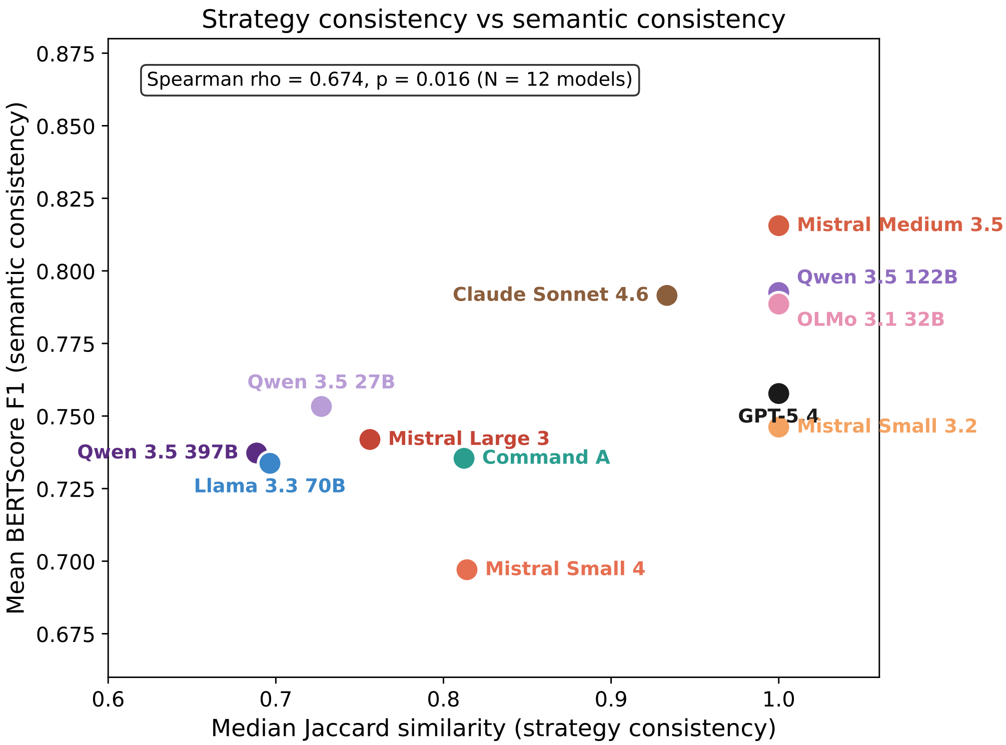


Figure 9: Plan consistency (median Jaccard) versus response similarity (mean BERTScore F1) per model, pooled across temperatures and vignettes.

across the sweep ($\rho = -0.66$), the steepest response-level temperature reaction among the top-ranked models, so temperature demonstrably reaches the model and it is the decisions that hold still while the wording drifts. Its consistency does not come from a narrow repertoire either: its pooled strategy distribution is broader than that of the proprietary models (59% top-two share), so its selections vary across patients but not across runs of the same patient.

The second addition behaves like an ordinary mid-field model, which is itself informative: Command A combines one of the broadest strategy repertoires in the set (47% top-two share) with mid-ranking decision consistency (median Jaccard 0.812), a genuine temperature response, and the framework's standard low-temperature optimum at $T$=0.15. The stricter collection route (pinned provider, enforced parameter forwarding) produces no qualitatively different behaviour from the original cohort.

The GPT-5.4 re-collection turns a measurement artefact into a finding. With temperature verifiably applied, GPT-5.4's mean decision consistency stays between 0.970 and 1.000 at every sampled temperature while its BERTScore declines from 0.770 to 0.745, the same decisions-stable-wording-drifts signature as Sonnet 4.6. The flat profile that the original collection produced for the wrong reason (the parameter was never applied) turns out to be genuine: GPT-5.4 is the most decision-stable model in the set (modal-set agreement 0.993), and its stability survives real sampling variance rather than reflecting its absence. At the trial level the two proprietary ceilings differ in kind. GPT-5.4 produces a single strategy set across all 20 trials in 28 of its 30 conditions, whereas Sonnet 4.6 produces two or three distinct sets in roughly half of its conditions at every temperature, including $T$=0.0. The two models are equally aligned (1.000) but differently deterministic, and Sonnet's non-monotonic median profile in Figure 6 reflects this temperature-independent baseline

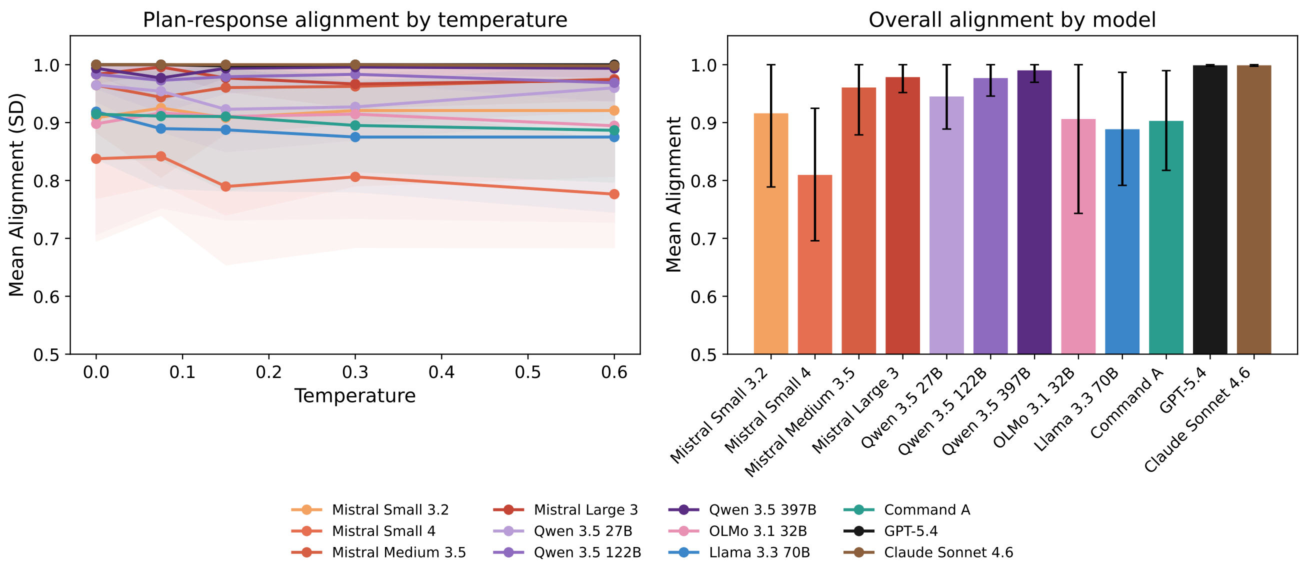


Figure 10: Plan–response alignment by model.

stochasticity passed through a median over six vignettes, not a temperature effect ($\rho = +0.03$, $p = .88$).

Each enlargement tightens the Bonferroni correction, from $\binom{10}{2} = 45$ to $\binom{11}{2} = 55$ and now $\binom{12}{2} = 66$ comparisons, moving two marginal pairs below significance in total (Jaccard: Llama 3.3 70B vs. Qwen 3.5 122B, $p_{\text{bonf}}$ 0.046 → 0.067; BERTScore: Mistral Small 3.2 vs. Sonnet 4.6, 0.035 → 0.051). No pairwise conclusion among unchanged models reversed direction and none became newly significant. All substantive findings of the original study hold under model-set enlargement and under the GPT-5.4 correction.

# 6 Discussion

## 6.1 What the Three Levels Reveal Together

The three methods were designed to capture distinct aspects of therapeutic stability, and the results confirm that they do: the cross-metric correlations are moderate at best. Each level reveals information the others miss, and the most informative finding comes from their separation.

Temperature reduces Jaccard but leaves alignment untouched. Models select different strategies as temperature rises, yet implement whichever strategy they select equally well. This pattern localises the source of instability to the *decision-making* layer rather than the *execution* layer. A model at high temperature does not produce incoherent or unfaithful responses. It produces coherent responses to a different therapeutic plan. The distinction matters in practice: if models selected consistent strategies but failed to carry them out, the remedy would lie in response generation and prompt design. The observed pattern is the opposite, so temperature and decoding policy are the more promising levers. It also matters diagnostically, because it means an evaluation that reads only the response text is looking at the wrong layer.

BERTScore illustrates the problem. Response similarity falls with temperature roughly twice as fast as decision consistency ($\rho = -0.49$ vs. $-0.22$), so wording varies before decisions do, and evaluations that compare only output text conflate harmless paraphrase with genuine decision churn.

In the other direction, different rescripting strategies share the same supportive framing and register, so a genuine strategy switch often leaves BERTScore unmoved. Neither is a failure of BERTScore itself, but a consequence of how therapeutic language is structured. The plan mechanism separates the two layers at the cost of one structured line per response, and the near-ceiling validity shows that every model can follow the format.

Finally, the near-ceiling alignment scores validate the framework's central simplification. Because plans are faithfully executed, Jaccard over declared strategy sets measures actual decision stability rather than label noise. Decision instability is real and model-specific: at validity 0.99+ and alignment near ceiling, the least stable models are not failing to follow the format or to implement their plans. They are changing their minds. In deployment terms, two patients, or the same patient after a page reload, can receive different therapeutic strategies for no clinical reason.

## 6.2 Cross-Model Comparison and EU-Sovereign Deployment

Does an EU-sovereign model achieve therapeutic stability comparable to proprietary alternatives? For the original study subject, the answer was a qualified yes. At its vendor-recommended temperature ($T$=0.15), Mistral Large 3 reaches a mean Jaccard of 0.848, against 1.000 for GPT-5.4 and 0.907 for Sonnet 4.6. The gap grows at higher temperatures, where Mistral Large 3 falls to 0.700 while the proprietary models stay at 0.87 or above.

The post-hoc addition changes the picture outright. Mistral Medium 3.5, open weights, EU-sovereign, dense 128B, is the most stable model in the entire set: perfect median decision consistency, the highest response-level consistency, and at or near ceiling at every sampled temperature. For deployments in which run-to-run consistency of clinical decisions is the binding constraint, these results support Medium 3.5 over Large 3 as the preferred EU-sovereign candidate. The scope of this claim is narrow: stability is necessary, not sufficient. This study measures neither therapeutic quality nor efficacy, and Medium 3.5's alignment (0.961) still trails the proprietary ceilings (1.000). The sovereignty conclusion nonetheless strengthens: at the time of the main collection the most stable models were mixed across origins, and with Medium 3.5 an EU-sovereign open-weight model tops the stability ranking.

Parameter count does not straightforwardly predict stability: Qwen 3.5 122B outperforms its 397B sibling at every temperature point, and the 24–32B dense models (Mistral Small 3.2, OLMo 3.1) sit at or near the top of the ranking at low temperatures. The 111–128B dense pair makes the point sharpest: Command A and Mistral Medium 3.5 are near neighbours in parameter count yet occupy the mid-field and the ceiling of the consistency ranking respectively, so the dense-architecture expectation that motivated the pair (Section 4.1) is not supported in its strong form: dense weights alone do not confer stability. The proprietary models show the flattest temperature profiles but are not in a class of their own. Part of the explanation is distributional, since models that concentrate on a narrow strategy repertoire have less room to vary (Section 5.1). Mistral Medium 3.5 is the counterexample that makes the ranking meaningful, combining a broad repertoire with perfect within-condition consistency: it adapts its strategy to the patient but not to the sampling seed.

## 6.3 Temperature and Clinical Deployment

The five-point sweep shows that the relationship between temperature and stability is not monotonic for all models. Ten of twelve models reach their best mean decision consistency at a non-zero temperature, and the entire Mistral family peaks between $T$=0.075 and $T$=0.15, exactly the region of its vendor default. Pushing a model to $T$=0.0 in the hope of maximising determinism can be counter-productive, and greedy decoding in any case does not deliver determinism: residual nondeterminism persists at $T$=0.0 (Yuan et al., 2025), and a fixed seed does not remove it (Appendix F).

The practical recommendation is to operate at or near the vendor-recommended temperature and to verify stability for the specific model at that setting. Models designed for a temperature range tend to be most stable within it, only three models show statistically significant degradation across the sweep (OLMo 3.1 32B, Qwen 3.5 122B, Qwen 3.5 27B), and vendor-recommended clinical temperatures differ per model. A deployment cannot simply "set temperature low" and assume stability, because the stable-at-any-temperature property itself separates models. Alignment softens the concern: the clinical cost of higher temperature is confined to strategy *selection*. A patient of a high-temperature model would encounter different therapeutic approaches across sessions, but each individual session would be internally coherent.

### 6.4 Robustness and Extensibility

Adding Mistral Medium 3.5 after the main data collection required a nine-line entry in the model registry (ID, temperature, reasoning mode) plus ten lines of figure display metadata; no metric, aggregation, or statistical-test code changed. The 600-trial collection ran unattended in 8 h 20 min of wall-clock time on the identical frozen histories. Command A later required the same single registry entry, and moving GPT-5.4 from the gateway to the native OpenAI API was a provider swap within its existing entry. The framework's extensibility is thus a demonstration rather than a claim, and the demonstration doubles as a robustness check: Section 5.7 shows all substantive findings survive both enlargements, with two marginal Bonferroni boundary crossings in total.

The GPT-5.4 correction carries a methodological lesson that extends beyond this study. An aggregation gateway can accept a sampling parameter and silently drop it: the request succeeds, the response is fluent, and nothing in any single output reveals that the sweep never varied. Only the stability measurement itself exposed the failure, as a BERTScore profile flat to within 0.006 while every genuinely responding model declined by 0.03 to 0.24 over the same sweep. Studies that manipulate sampling parameters through an intermediary should verify end to end that the parameters reach the model, pin the serving provider, and configure requests to fail rather than proceed when an endpoint cannot honour a parameter. This study's later collections adopt exactly that policy (Section 4.1).

### 6.5 Framework Reflections

The consistency scores depend on how strategies are categorised. Broader categories inflate Jaccard because more selections land in the same bin, finer categories deflate it. The six-category taxonomy balances clinical detail against discriminability, but different category boundaries would shift the absolute numbers. The relative ordering of models is more robust, since all models are evaluated against the same set.

The plan block also has limits. It captures what the model declares, not what it internally computes, and it cannot tell apart a model that picks a strategy and then writes the response from one that writes a response and then picks a matching label. The high alignment scores fit either reading. What matters for the clinical argument is that declarations vary across runs, and that variation predicts variable therapeutic behaviour regardless of the underlying process.

Beyond IRT, the framework carries over to any structured protocol where the therapist chooses from a defined set of techniques. Cognitive behavioural therapy and motivational interviewing are natural candidates. Each would need its own strategy taxonomy, but the evaluation setup and metrics transfer unchanged.

### 6.6 Limitations

All patient interactions are simulated. Synthetic patients follow scripted behavioural profiles and do not exhibit the unpredictable responses of real patients, and six profiles sample the difficulty range without exhausting it. No human participants were involved, so the results measure computational consistency rather than clinical safety or efficacy. IRT targets nightmares rather than psychiatric diagnoses, which limits clinical risk, but any deployment to real patients would require validation beyond what this framework provides.

The study evaluates a single phase (rescripting) of a single protocol, at a single conversation depth (slice 2, chosen as a conservative lower bound). Stability in other IRT stages, at other depths, or in other protocols may differ.

The measurement itself carries residual variance from the serving side. The original collections used OpenRouter's default routing, under which the third-party serving stack (provider, quantisation, batching) may differ between requests; the later collections pin one provider per model and log it per trial, but the two regimes are not identical. The judge endpoints are versioned previews whose revision may likewise differ between collections, and the human validation ($\kappa = 0.777$) covers 60 trials, below 1% of the total. Judge reliability at scale remains an open question (Zheng et al., 2023).

Finally, the taxonomy constrains what can be measured, and the statistics are exploratory. Strategies outside the six categories are invisible to Method 1, different category boundaries would produce different absolute scores, and with 30 runs per model the Bonferroni-corrected pairwise tests are conservative.

## 7 Conclusion

This paper asked how consistent LLM therapeutic responses are across independent runs within a structured IRT protocol. Three evaluation methods were applied to twelve models across 7,200 trials spanning five temperature conditions, on frozen histories that make every trial's input byte-identical.

The main findings are:

1. Consistency varies widely across models and far less across patients. Median decision consistency spans 0.69 to 1.00 between models, while the vignette effect is not significant. Stability is a model property.

2. The instability sits in strategy selection, not execution. Temperature reduces plan consistency but not plan–response alignment: models pick different strategies at higher temperatures but carry out whichever strategy they pick equally well.

3. Greedy decoding is not always optimal. Ten of twelve models reach their best mean consistency at a low non-zero temperature, and the Mistral family peaks at its vendor default. Deployments should verify stability at the vendor-recommended setting rather than assume $T$=0.0 is safest.

4. No single metric captures the full picture. The three levels correlate only moderately ($\rho = 0.29$ to 0.55). BERTScore misses strategy changes and overstates drift from harmless paraphrase, and only the combination localises where instability lives.

5. The framework is cheap to extend, and the findings are robust to model-set changes. Adding a model requires a single configuration entry on identical frozen inputs, the framework's own stability signature exposed a gateway silently dropping GPT-5.4's temperature parameter, and every substantive finding holds across these revisions. Mistral Medium 3.5, an EU-sovereign open-weight model, tops the overall stability ranking.

The three-level framework fulfilled its design goal by identifying that instability concentrates at the strategy selection layer. Neither BERTScore nor alignment alone would have revealed that. The cross-model comparison shows that EU-sovereign deployment no longer requires a stability compromise: what was a small gap for Mistral Large 3 at its recommended temperature became, with Mistral Medium 3.5, the top of the ranking.

Several directions follow. Extending the evaluation beyond the rescripting phase would test whether the patterns hold across different types of clinical decisions, and the framework transfers to other manualised protocols (CBT, motivational interviewing) given a protocol-specific taxonomy. Since vendors update weights and serving infrastructure without notice, repeating the evaluation across model versions would show whether stability improves or fluctuates over release cycles. reDreamAI targets German-speaking users, so evaluating stability in German would test whether the English-language patterns transfer. A larger human annotation study would strengthen the judge validation beyond the current 60-trial subset. Finally, because consistency is necessary but not sufficient, the natural companion to this framework is an evaluation of therapeutic quality: a model must make the same decision every time, and it must be the right one.

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| Category | Mechanism |
|---|---|
| `confrontation` | External action toward the threat (fight, challenge, resist) |
| `self_empowerment` | Internal transformation (grow stronger, gain abilities) |
| `safety` | Environment modification (safe spaces, protective elements) |
| `cognitive_reframe` | Meaning change (reinterpret threatening elements) |
| `social_support` | Adding allies (introduce helpful figures) |
| `sensory_modulation` | Sensory and calming details (modify light, sound, texture; introduce comforting elements) |

Table 2: Final strategy taxonomy (v4).

# A Strategy Taxonomy

The six mechanism-level categories that constrain plan declarations:

The taxonomy evolved over four iterations ($8 \to 6 \to 7 \to 6$ categories). The initial version suffered a severe skew: a broad empowerment/mastery category absorbed 87.7% of all picks. Merging it made the skew worse (100% dominance of a single "agency" category); the resolution was a mechanism-level split distinguishing external action (`confrontation`) from internal transformation (`self_empowerment`). A seventh category (`emotional_regulation`) was merged into `sensory_modulation` after smoke testing showed it was picked in 0.4% of trials. Category definitions describe *how* the dream changes (mechanism), not what the change achieves (outcome), with NOT-examples to reduce overlap.

# B Prompts

The fused-generation system prompt injected into each trial (the `{categories_block}` placeholder is filled with the six taxonomy definitions of Appendix A):

```
You are an Imagery Rehearsal Therapy (IRT) assistant helping a patient rescript their
nightmare.
First, select 1-2 therapeutic strategies from this taxonomy:
{categories_block}
Choose the strategies most specifically suited to THIS patient's nightmare content.
Avoid defaulting to general approaches when a more specific mechanism fits.
Then respond to the patient using those strategies.
STRICT output format:
Line 1:  <plan>category1 / category2</plan>
Line 2:  (empty)
Line 3+:  Your therapeutic response (conversational, supportive, max 3 sentences)
IMPORTANT:
- You MUST include the closing </plan> tag on the SAME line as the opening tag.
- The <plan> line must contain ONLY category names, nothing else.
- Do NOT put your response inside the <plan> tags.
```

An in-context example provided with the prompt:

```
<plan>safety / social_support</plan>
What if, in this version of the dream, you could find a place that feels completely
secure -- and someone you trust is already there waiting for you?
```

Each trial receives this system prompt together with the frozen dialogue history (slice 2 of the vignette's generated session), so all trials of a condition share byte-identical input.

## C Judge Rubric

The alignment judge receives the taxonomy definitions, the trial's declared strategies, and the therapist response, and scores each declared strategy on a ternary scale with mandatory reasoning:

```
For each declared strategy, assign ONE score:
- 2 (implemented):  The response clearly and specifically implements this strategy.
The therapeutic technique is evident and developed -- you can point to specific sentences
or passages that demonstrate it.
- 1 (partial):  The response touches on this strategy but does not fully develop
it.  Elements are present but incomplete, only implied, or mentioned without therapeutic
depth.
- 0 (absent):  The response does not implement this strategy.  No evidence of this
technique in the response.
Output format:  strategy_id:  <reasoning> | score:  <0|1|2>
```

Per-trial alignment is the mean score over declared strategies, normalised by the maximum (2). Trials whose judgment fails (timeout, API error) or that cannot be judged (invalid plan) are treated as missing data and excluded from the alignment mean. Zero-scored or partially parsed judgments are re-judged once by Gemini 3.1 Pro (9 of 7,200 trials; Section 5.6).

## D Plan Validity

Figure 11 reports the plan validity rate per model, the upstream quality gate for Method 1.

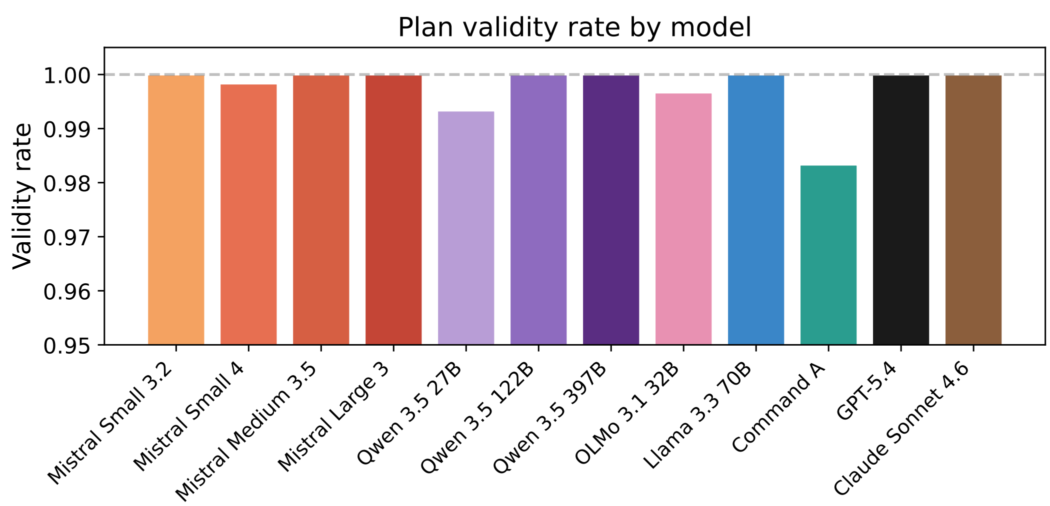


Figure 11: Plan validity rate by model: fraction of trials whose `<plan>` block parses to valid taxonomy categories.

# E Per-Vignette Stability

Figure 12 breaks decision consistency down to individual model–vignette cells, one heatmap per temperature, for consultation of specific combinations.

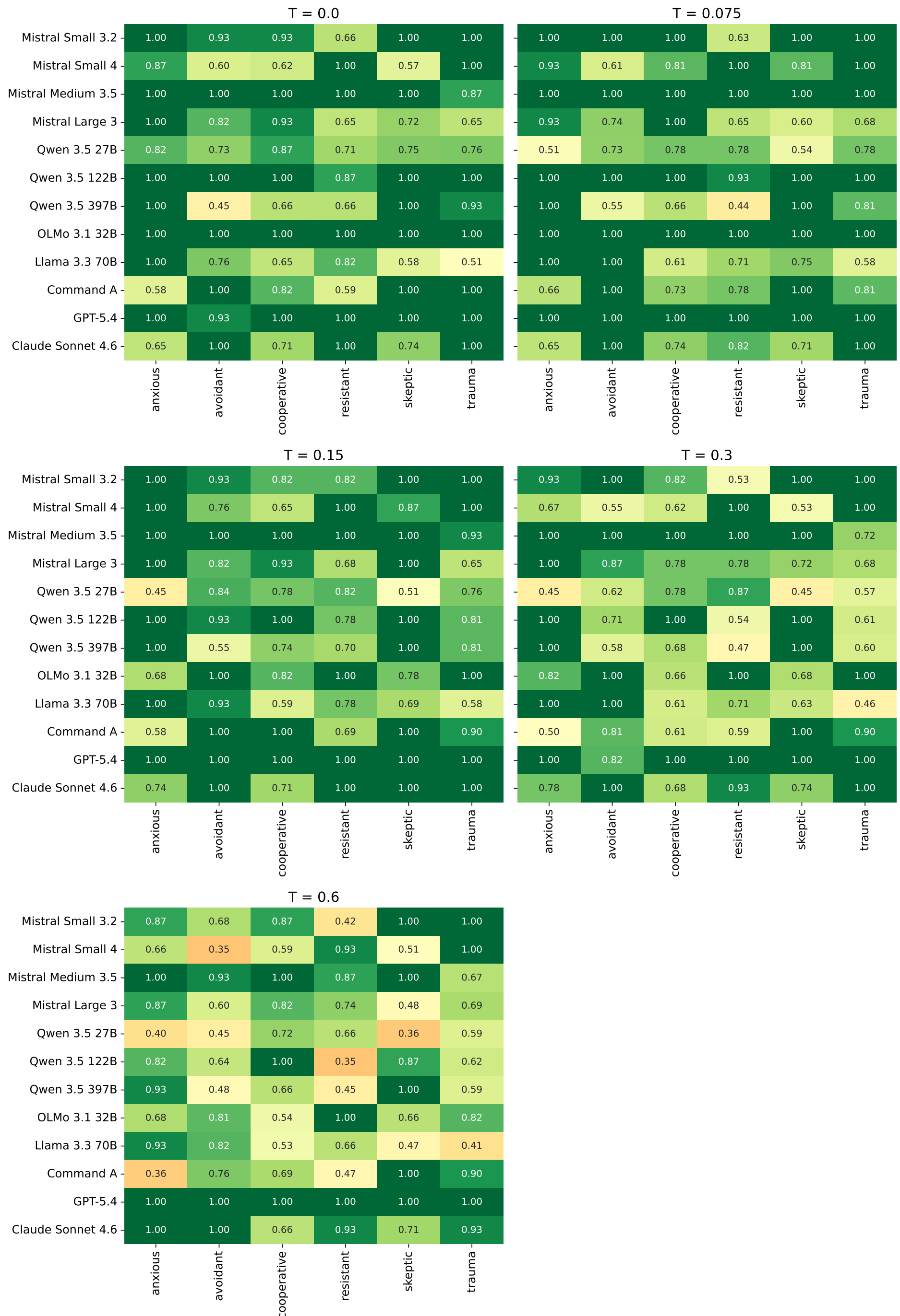

Figure 12: Median Jaccard per model × vignette, one heatmap per temperature. Individual model–vignette combinations can deviate noticeably from the aggregate even though the overall vignette effect is not significant (Section 5.4).

# F Seed Control Experiment

A supplementary experiment on Mistral Large 3 compares runs with a fixed seed (`seed=42`) against unseeded runs under otherwise identical conditions. Fixing the seed does not improve consistency (Figure 13): vendor seed support is best effort at most, and residual serving-side nondeterminism persists (Yuan et al., 2025). The main experiment therefore sets no seed and captures stochastic variance through trial repetition instead.

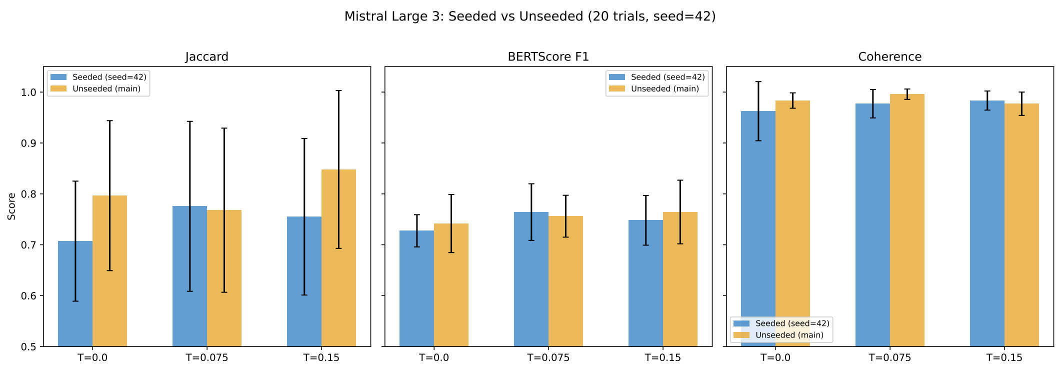


Figure 13: Seeded (`seed=42`) versus unseeded runs for Mistral Large 3. A fixed seed does not improve run-to-run consistency.

## G Per-Condition Results

| Model | $T$=0.0 | $T$=0.075 | $T$=0.15 | $T$=0.3 | $T$=0.6 |
|---|---|---|---|---|---|
| Mistral Small 3.2 | 0.967 | 1.000 | 0.967 | 0.967 | 0.872 |
| Mistral Small 4 | 0.746 | 0.874 | 0.935 | 0.646 | 0.625 |
| Mistral Medium 3.5 | 1.000 | 1.000 | 1.000 | 1.000 | 0.967 |
| Mistral Large 3 | 0.772 | 0.709 | 0.877 | 0.775 | 0.712 |
| Qwen 3.5 27B | 0.760 | 0.754 | 0.768 | 0.595 | 0.518 |
| Qwen 3.5 122B | 1.000 | 1.000 | 0.967 | 0.853 | 0.730 |
| Qwen 3.5 397B | 0.799 | 0.737 | 0.775 | 0.642 | 0.626 |
| OLMo 3.1 32B | 1.000 | 1.000 | 0.911 | 0.911 | 0.746 |
| Llama 3.3 70B | 0.705 | 0.730 | 0.732 | 0.668 | 0.597 |
| Command A | 0.911 | 0.793 | 0.950 | 0.714 | 0.724 |
| GPT-5.4 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Claude Sonnet 4.6 | 0.868 | 0.779 | 1.000 | 0.854 | 0.933 |

Table 3: Median pairwise Jaccard per model and temperature (across 6 vignettes).

| Model | $T$=0.0 | $T$=0.075 | $T$=0.15 | $T$=0.3 | $T$=0.6 |
|---|---|---|---|---|---|
| Mistral Small 3.2 | 0.757 | 0.766 | 0.758 | 0.743 | 0.708 |
| Mistral Small 4 | 0.692 | 0.725 | 0.726 | 0.684 | 0.657 |
| Mistral Medium 3.5 | 0.830 | 0.886 | 0.824 | 0.785 | 0.754 |
| Mistral Large 3 | 0.742 | 0.756 | 0.764 | 0.743 | 0.705 |
| Qwen 3.5 27B | 0.831 | 0.780 | 0.754 | 0.715 | 0.687 |
| Qwen 3.5 122B | 0.882 | 0.840 | 0.789 | 0.743 | 0.710 |
| Qwen 3.5 397B | 0.750 | 0.759 | 0.735 | 0.726 | 0.717 |
| OLMo 3.1 32B | 0.941 | 0.813 | 0.760 | 0.730 | 0.699 |
| Llama 3.3 70B | 0.746 | 0.749 | 0.740 | 0.731 | 0.702 |
| Command A | 0.771 | 0.762 | 0.754 | 0.700 | 0.690 |
| GPT-5.4 | 0.770 | 0.762 | 0.760 | 0.751 | 0.745 |
| Claude Sonnet 4.6 | 0.844 | 0.801 | 0.799 | 0.771 | 0.743 |

Table 4: Mean pairwise BERTScore F1 per model and temperature (across 6 vignettes).

| Model | $T$=0.0 | $T$=0.075 | $T$=0.15 | $T$=0.3 | $T$=0.6 |
|---|---|---|---|---|---|
| Mistral Small 3.2 | 0.908 | 0.925 | 0.908 | 0.921 | 0.921 |
| Mistral Small 4 | 0.837 | 0.842 | 0.790 | 0.806 | 0.776 |
| Mistral Medium 3.5 | 0.965 | 0.944 | 0.960 | 0.963 | 0.975 |
| Mistral Large 3 | 0.983 | 0.996 | 0.977 | 0.967 | 0.973 |
| Qwen 3.5 27B | 0.965 | 0.954 | 0.923 | 0.927 | 0.960 |
| Qwen 3.5 122B | 0.983 | 0.973 | 0.979 | 0.983 | 0.969 |
| Qwen 3.5 397B | 0.994 | 0.977 | 0.994 | 0.996 | 0.994 |
| OLMo 3.1 32B | 0.898 | 0.917 | 0.910 | 0.915 | 0.894 |
| Llama 3.3 70B | 0.919 | 0.890 | 0.888 | 0.875 | 0.875 |
| Command A | 0.915 | 0.911 | 0.910 | 0.895 | 0.887 |
| GPT-5.4 | 1.000 | 1.000 | 0.998 | 1.000 | 1.000 |
| Claude Sonnet 4.6 | 1.000 | 1.000 | 1.000 | 1.000 | 0.998 |

Table 5: Mean plan–response alignment per model and temperature (across 6 vignettes).